

NEXCORT iQ · INTERNAL PRODUCT REQUIREMENTS

Data Flow & Architecture Overview

Plain-language technical overview for agency IT, procurement, and pilot stakeholders.

Legal entity: Apps on Demand LLC (Wyoming) · Product: NexCort iQ · Marketing: www.nexcortiq.us · Ops: app.rapidcortex.us · (404) 520-1747 · jeff@nexcortiq.us

High-level architecture

- Marketing: www.nexcortiq.us. Authenticated ops console: app.rapidcortex.us (Next.js on ECS Fargate + CloudFront/WAF).
- Identity: Amazon Cognito JWT (custom:role, custom:agencyId). Product vertical roles are preserved before any PSAP role migration.
- APIs: API Gateway HTTP APIs + AWS Lambda (Node.js 22 / Python 3.11). Nested SAM stacks split by domain (incidents/CAD, language/analytics, media/admin, billing, QR, Vision).
- Data: DynamoDB (agency-scoped keys), S3 for media, KVS for live video, CloudWatch/CloudTrail for ops evidence.
- AI: Amazon Bedrock, Transcribe, Translate — assistive only; mock paths exist for non-prod.
- CAD: read-only adapters in pilot. Write-back is a separate, fail-closed path.

Environment note

Live production web is app.rapidcortex.us. Engineering staging is isolated (app-staging.rapidcortex.us). CloudFormation DeploymentStage=dev maps to live production — it is not a sandbox. Never attach staging API custom domains to api.rapidcortex.us.

Caller media flow

Step	What happens	Control
1	Authorized dispatcher/security role requests media during an incident.	RBAC + agency check
2	System sends a short-lived SMS/web link tied to the incident/session.	Expiring token; 10DLC SMS
3	Caller uploads photo/video or starts WebRTC Video Assist (no app install).	No public buckets; encrypted object
4	Console displays media to authorized roles.	Access audited (user, time, incident)
5	Retention policy expires or preserves content per agency config.	Pilot retention rules

Incident intelligence flow

Input	Processing	Output	Human control
Live audio / notes / media	Transcription, translation, SOP/AI coaching, confidence cues	Transcript, summary, protocol reminders	Dispatcher/supervisor review before use
CAD read-only	Adapter normalizes non-sensitive fields	Context panel in NexCort iQ	No write-back in default pilot
Vision cameras (if in SOW)	Authorized session; observations + vision transcripts	Console Vision events (rapid-vision.*)	Role-gated; not autonomous enforcement
Audit events	Append-style logging via AuditRepository	Agency audit trail / export	Admin/export controls; audit failure must not abort ops

Product surfaces (do not mix consoles)

Surface	Who	Where
NC 911 dispatcher	dispatcher (and supervisor monitoring)	/jurisdiction}/dashboard — not for agencyadmin
Supervisor / QA	supervisor	/jurisdiction}/supervisor
Agency admin / IT	agencyadmin, agencyit	/jurisdiction}/admin and /admin/it
Campus / venue / transit / hospital	Vertical roles only	/app/campus venue transit hospital-* — no PSAP CAD queue
Call Assist	CALL_ASSIST_*	/app/call-assist/* — never 911 dispatcher workspace
Sales contractor	salescontractor	/sales + curated RC Admin CRM/intel. No tenant/billing/flags.
RC Admin	rcsuperadmin, radmin, rcitadmin	/rc-admin/* (IT lands on infrastructure)

Production architecture decision

Separated engineering vs live environments; CloudFront + WAF at the edge; ECS/Fargate for the web app; nested SAM stacks under the 200 KB / transform-size limits; automated health checks and rollback. New Lambdas go in the correct domain stack — do not bloat the primary template.